Gdańsk University of Technology
Faculty of Electronics, Telecommunications and Informatics
Department of Multimedia Systems

Technical Report

AI - Optic Disc & Cup Segmentation

v1.0

Proposed Improvements to the Glaucoma Segmentation Model (U-Net++)

Project Subject: Glaucoma detection using intelligent analysis of fundus images

Team Members: Martyna Borkowska (Lead), Karolina Glaza, Mateusz Dobry, Agnieszka Pawłowska, Amila Amarasekara, Antoni Naczke

Supervisor: Prof. Andrzej Czyżewski, Ph.D., D.Sc., Eng.

Authors: Mateusz Dobry

2026

Contents

1 Introduction

Goal: increase the Dice score from ∼90% to ≥95%.
Dataset: 1020 images (target ∼5820).

This document lists proposed improvements to the U-Net++ segmentation model, grouped by priority and by the effort required, together with the estimated impact on the Dice score and the implementation difficulty.

2 Critical Improvements - Immediate Implementation

Change	**Why**	**Est. impact**	**Difficulty**
Fix the `val_dataset` transform bug	The current code overrides the transforms for the whole dataset (including training), so validation does not work correctly. Fixing it yields reliable metrics and improves training quality.	+1-3%	Easy
Dice + Focal Loss or Tversky Loss	Focal Loss focuses on hard pixels (disc/cup boundaries). Tversky lets you weight FP and FN separately - clinically important (missing glaucoma costs more than a false alarm).	+1-3%	Easy
Mixed Precision Training (AMP)	float16/float32 training allows raising the batch size from 8 to 16-24. Larger batches stabilise Batch Normalization in the encoder - a direct Dice increase.	+1-2%	Easy

3 Important Improvements - New Training / Inference

4 Additional Improvements - After Dataset Expansion

5 Combined Potential

Implementing the improvements from Sections 2 and 3 should be enough to reach ≥95% Dice. Section 4 becomes relevant after the dataset is expanded to ∼5820 images.

6 Summary

The critical fixes (validation transform, loss function, AMP) are low-effort, high-value and should be applied immediately. Combined with the important training/inference changes,

Change	**Why**	**Est. impact**	**Difficulty**
Test-Time Augmentation (TTA)	Predict on several versions of the image (flip, rotation) and average. No retraining - removes random segmentation artefacts; results are more consistent and reproducible.	+0.5-1.5%	Easy
CLAHE + fundus-specific augmentations	CLAHE is a standard preprocessing step in ophthalmology - it equalises contrast locally and sharpens disc boundaries. RandomGamma simulates different fundus cameras. Crucial for multi-clinic data.	+1-2%	Medium
Cosine Annealing + Linear Warmup	The current ReduceLROnPlateau is reactive. Warmup protects the encoder's pretrained weights; the cosine LR decay improves the model's final fit.	+0.5-1%	Easy
K-Fold CV (5 folds) + Ensemble	With 1020 images a single split is too little (∼200 val). 5-Fold gives a reliable estimate plus 5 models for an ensemble, which when averaged consistently beats a single model.	+1-2%	Medium

they are expected to reach the ≥95% Dice target on the current dataset; the additional changes pay off once the dataset is expanded.

Change	**Why**	**Est. impact**	**Difficulty**
Morphological post-processing	Closing fills holes in the masks; Connected Components removes artefacts. Anatomical constraint: the cup must lie inside the disc - this can be enforced programmatically.	+0.5-1%	Medium
Change the encoder (EfficientNet-B5 / ConvNeXt-Base / MiT-B3)	With 5820 images a larger/newer encoder will no longer overfit. MiT-B3 (SegFormer) is designed specifically for medical segmentation.	+0.5-2%	Easy

Stage	**Total estimated gain**
Section 2 - Critical (1020 images)	+3-8% Dice
Sections 2 + 3 - Critical + Important (1020 images)	+5-10% Dice $\rightarrow$ $\geq$95% achievable
Sections 2 + 3 + 4 (5820 images)	+6-13% Dice $\rightarrow$ full optimisation